RAJALAKSHMI ENGINEERING COLLEGE

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Course Code	**AD23532**	Course Name	**Principles of Data Science**				
Course Type	**Lab Oriented Theory**	Course Category	**Professional Core (PC)**	**L**	**T**	**P**	**C**
				2	**0**	**4**	**4**
Regulation	**2023**		Academic Year	**2026-27 (ODD)**			
Degree and Branch	**B.Tech. Artificial Intelligence and Data Science**		Batch	**2024-2028**			
Submission Date:		**03-10-2026**	Semester:	**V**			

Review – 3
Model Building and Comparison Analysis

In the last phase of the mini project, focus on model building and performance comparison. Develop and train suitable machine learning models for your selected dataset. Evaluate and compare all the models using appropriate performance metrics relevant to your problem statement (e.g., accuracy, precision, recall, F1-score, silhouette score, etc.).

Provide a clear analysis and discussion on how each model performs and identify which model gives the best results. Finally, present a better solution for your problem by justifying why the selected model outperforms the others, supported by evidence from your evaluation metrics and reasoning.

Expected Deliverables

1. Model Selection and Justification
   - Select at least two or more suitable machine learning algorithms based on the problem statement.
   - Justify the choice of algorithms.
2. Model Development
   - Train and validate the selected machine learning models.
   - Apply appropriate train-test split or cross-validation techniques.
3. Model Evaluation
   - Evaluate each model using appropriate performance metrics.
   - Use suitable evaluation methods based on the problem type:
     - Classification: Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix
     - Regression: MAE, MSE, RMSE, R^2 Score
     - Clustering: Silhouette Score, Davies-Bouldin Index, Calinski-Harabasz Score
4. Model Performance Comparison
   - Compare all developed models using tables, charts, and performance metrics.

    - Analyze the strengths and limitations of each model.
5. Hyperparameter Tuning and Model Optimization (if applicable)
    - Improve model performance using techniques such as Grid Search, Random Search, or Cross-Validation.
    - Compare model performance before and after optimization.
6. Final Model Selection
    - Identify the best-performing model.
    - Justify the selection with quantitative results and supporting analysis.
7. Updated Project Documentation
    - Include model implementation, evaluation results, comparison, visualizations, and conclusions.
    - Submit well-documented code/notebook and technical report.

Evaluation Criteria

Criteria	**Marks**
Model implementation and training methodology	10
Appropriate use of evaluation metrics	5
Performance comparison and analytical discussion	5
Hyperparameter tuning and model optimization	5
End-to-end product development and functionality	10
Technical report	5
Total	**40**